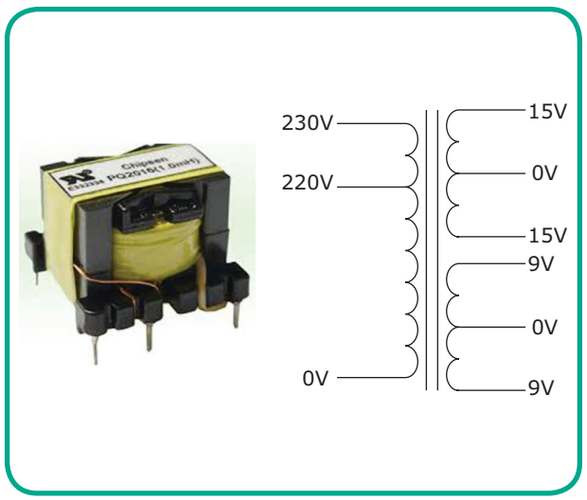


Figure 2.23 Power Transformer

DO YOU KNOW?

What is the difference between electrical and electronics devices?

The devices that work on AC (Alternating Current) are called Electrical devices and the ones that work on DC (Direct Current) are called Electronic devices.

2.3.4 Transformer Losses

Even in best transformers, unavoidable losses are occurred. These losses may be reduced using quality materials and cannot be totally nullified. The losses are of three types.

Copper Loss: The loss appeared because of the resistivity of copper string in the coil.

Hysteresis Loss: When AC changes its phase, the magnetic phase also gets changed. This causes some loss in the strength of the current. This is known as Hysteresis loss.

Eddy Current Loss: When current flows in an iron, it will get heated. This heat creates some loss. This loss is said to be Eddy current loss. To minimize this loss, the core should be laminated.

2.4. MICROPHONES AND LOUD SPEAKERS

In this Section, we are going to study about Microphones and loud speakers. They may also be called as transducers, because they are converting one form of energy into another form.

2.4.1 Microphone

A device, which converts sound waves into electrical waves, is called as microphone. There are few types based on the construction.

Types of microphone

1. Carbon microphone
2. Ribbon microphone
3. Dynamic microphone
4. Condenser microphone

Let us see some of the important microphones.

2.4.1.1 Dynamic Microphone

The dynamic microphone can also be called as moving coil microphone, which is shown in Figure 2.24. It is working under the principle of electromagnetic induction. This works on the basis of the following principle which states that "in a magnetic field a conductor is placed in such a way that it cuts the magnetic field and then inducing an electrical field in it".

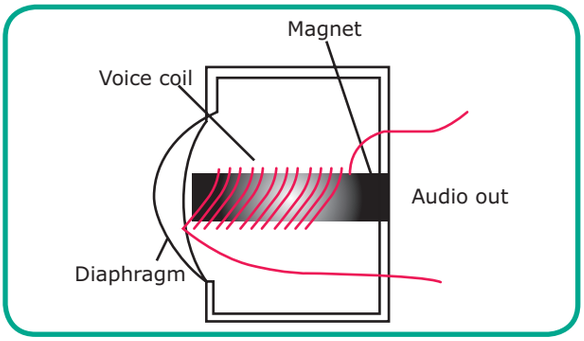


Figure 2.24 Dynamic Microphone

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